

LSTN: A Linear Model of Industrial Production Process for Demand Response

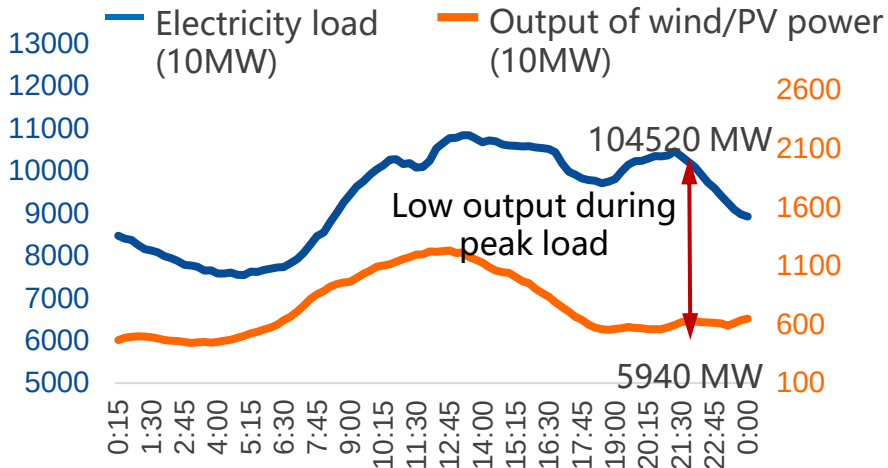
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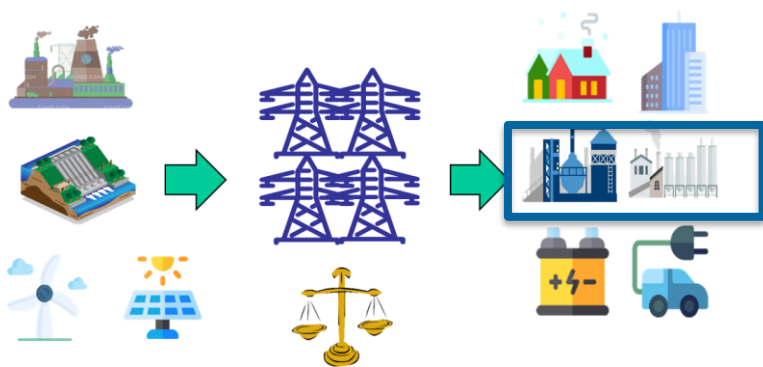
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Introduction



Electricity load and wind/PV power output curve on Aug 3, 2020, Jiangsu Province, China



Demand response (DR) has huge potential for maintaining demand-supply balance

Why demand response?

- **Cost-effective** for securing electricity supply in the power system

Why industrial users?

- **Large** response volume (MWs)
- **Sensitive** to the prices
- **Ability** to implement DR

Introduction

Industrial Production Processes (IPPs).

- **Contributes most energy consumption** of typical industrial users
- Exists technical constraints and production targets
- Modelled by: state-task network (STN), resource-task network...
- **MILP formulations** for the on/off operating states of devices

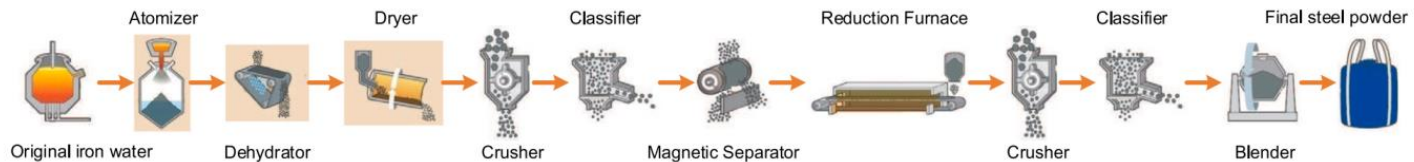


Fig. 4. The production process of a steel powder manufacturing facility.

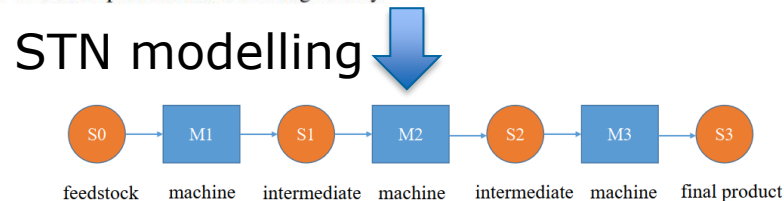


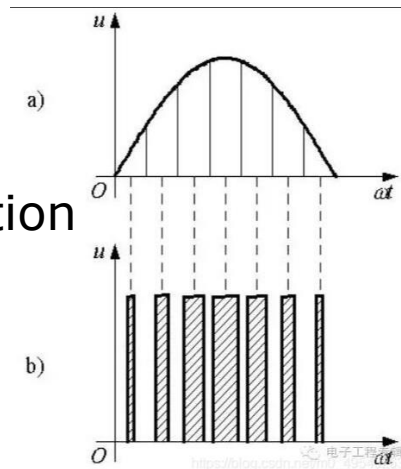
Fig. 1. A common state-task network model for industrial production process.

Objectives

Linear model instead of MILP-based models

- Overcome computation complexity issues
- Increase production model accuracy
- Easier to embed into existing models

An analogy:
PWM rectification
(Pulse Width
Modulation)



Power is discrete, but the energy integrated over a time interval can be close to continuous

$$\begin{aligned} &\min/\max c(\mathbf{x}) \\ &\text{s.t. } f(\mathbf{x}) \leq 0 \\ &\quad g(\mathbf{x}) = 0 \\ &\quad \mathbf{x} \in \Omega \end{aligned}$$

Linear constraints are more “friendly” for power system dispatch models

Methodology

Use **operating time** as the variable

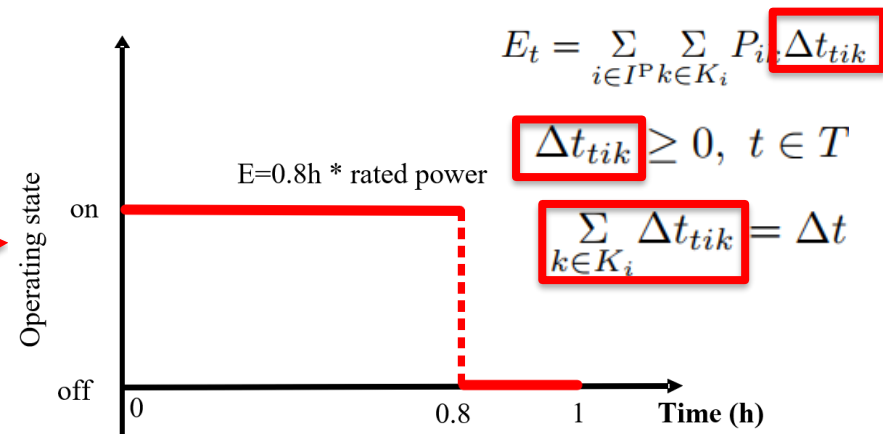
- Under the time intervals of DR (e.g. hour by hour)
- Change the variable: device operating state -> operating time
- Feasible; up to the industrial user to implement

STN

hour	1	2	3	...
state of device 1	on	on	off	...



hour	1	2	3	...
on-time of device 1	1h	0.8h	0h	...
off-time of device 1	0h	0.2h	1h	



The industrial user implements the operation of the devices

LSTN: linear STN

Methodology

Use **operating time** as the variable

$$Cost = \sum_{t \in T} Pr_t E_t, \quad (1)$$

$$E_t = \sum_{i \in I^P} \sum_{k \in K_i} P_{ik} \Delta t_{tik}, \quad t \in T, \quad (2)$$

$$\Delta t_{tik} \geq 0, \quad t \in T, i \in I^P, k \in K_i. \quad (3)$$

$$\sum_{k \in K_i} \Delta t_{tik} = \Delta t, \quad t \in T, i \in I^P. \quad (4)$$

$$S_{ti} \geq S_i^0 + S_i^{\text{tar}}, \quad t = t^{\text{end}}, i \in I^S. \quad (5)$$

$$0 \leq S_{ti} \leq S_i^{\text{max}}, \quad t \in T, i \in I^S. \quad (6)$$

$$S_{ti} = S_i^0, \quad i \in I^S, t = 0 \quad (7)$$

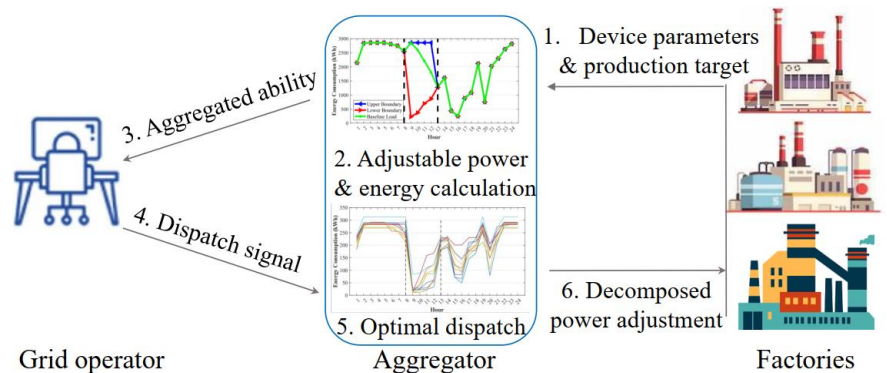
$$S_{ti} = S_{(t-1)i} - \sum_{k \in K_{i+1}} C_{(i+1)k} \Delta t_{t(i+1)k}, \quad t \in T, i = 0. \quad (8)$$

$$S_{ti} = S_{d(t-1)i} + \sum_{k \in K_i} G_{ik} \Delta t_{tik} - \sum_{k \in K_{i+1}} C_{(i+1)k} \Delta t_{t(i+1)k}, \quad t \in T, i \in I^P \setminus \{i^{\text{end}}\}. \quad (9)$$

$$S_{ti} = S_{d(t-1)i} + \sum_{k \in K_i} G_{ik} \Delta t_{tik}, \quad t \in T, i = i^{\text{end}}. \quad (10)$$

No need to worry
about the equations :)

- A linear model
- Fast to compute
- Suits aggregation
- Accuracy?



A two-layer framework for aggregating industrial facilities for large-scale DR

Case Study

- Tested using electricity price of PJM and parameters of a steel powder manufacturing.
- “True” value: STN-2min (it takes 2 min to switch)

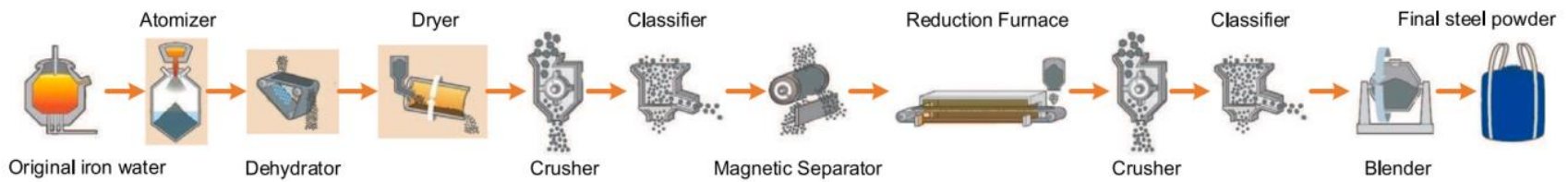
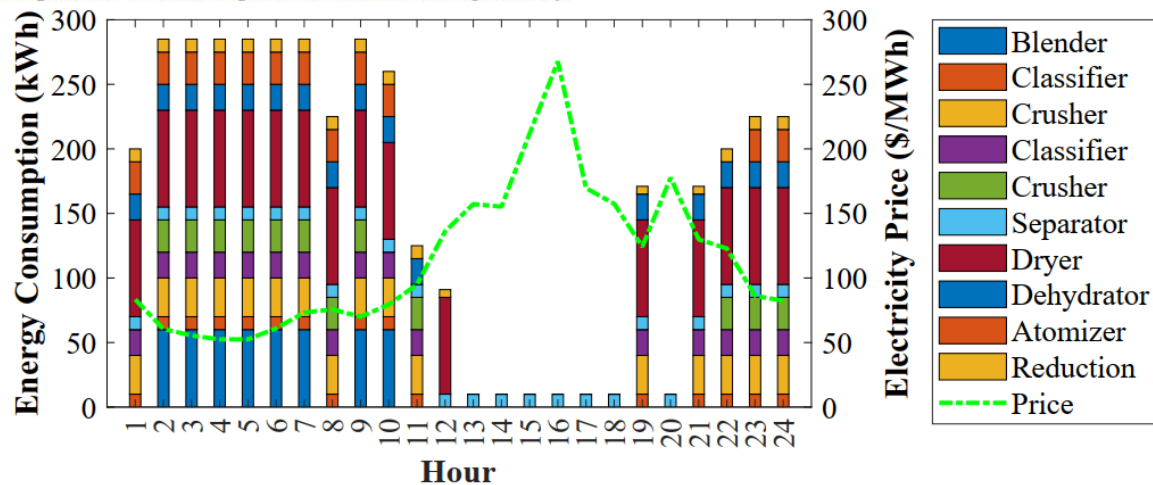


Fig. 4. The production process of a steel powder manufacturing facility.

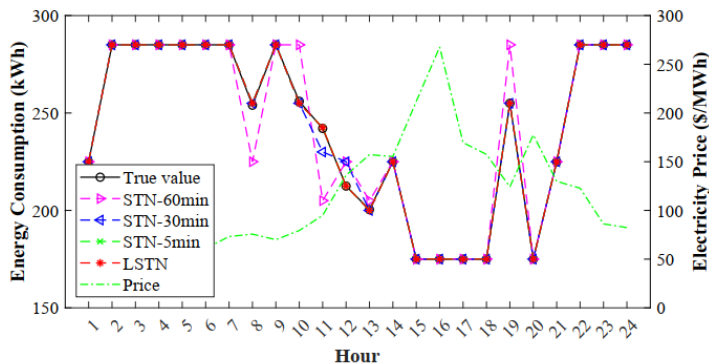


Results

Modeling Accuracy

TABLE II
THE PERFORMANCE OF THE COMPARED MODELS.

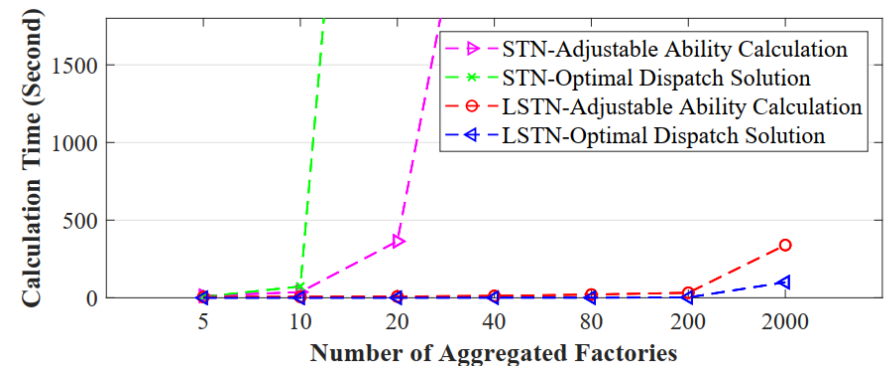
Models	RMSE for load profile (kW) /Ratio to max. value	RMSE for energy cost (\$) /Ratio to max. value
STN-60min	15.3(5.37%)	1.979(0.175%)
STN-30min	3.43(1.21%)	0.398(0.035%)
STN-5min	0.59(0.21%)	0.034(0.003%)
LSTN	0.30(0.17%)	0.033(0.003%)



LSTN: red **True: black**

Fig. 3. Load profile on August 5 given by the compared models.

Computational Time



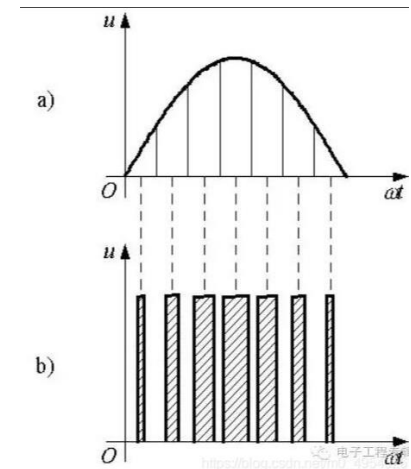
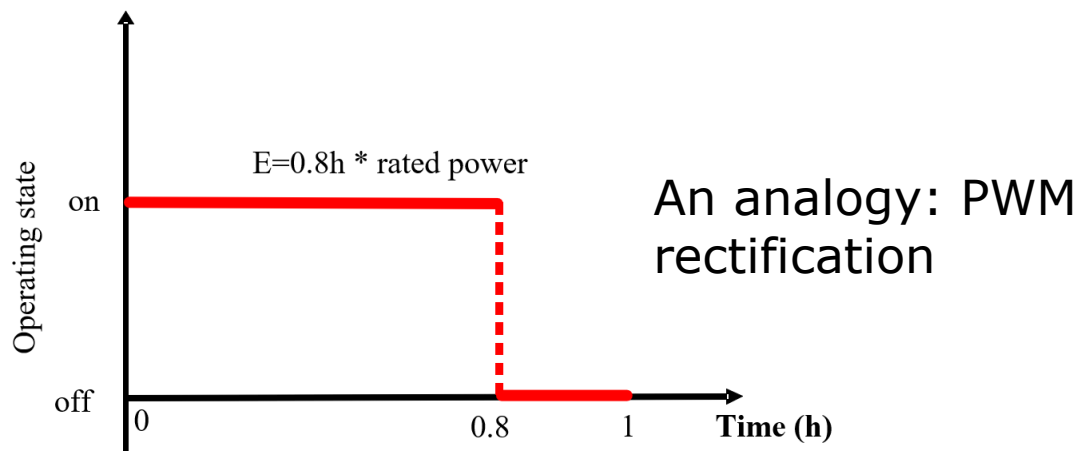
STN-60min: Computationally intractable with more than 80 industrial facilities

LSTN: Computationally tractable with 2000 industrial facilities or more

Conclusions

LSTN: Linear STN instead of MILP-based models

- Overcome computation complexity issues
- Increased model accuracy
- Easier to embed into dispatch models



Power is discrete, but the energy integrated over a time interval can be close to continuous!